LLMs for Text Analysis

Using Large Language Models in Social Science Research

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What Are LLMs?

Definition

Large Language Models are neural networks trained on massive text corpora, capable of understanding and generating human language.

Proprietary (API access)

- GPT-4 (OpenAI)
- Claude (Anthropic)
- Gemini (Google)

Pay per use; hosted remotely; no model weights shared.

Open-weight (run locally)

- Llama (Meta)
- Mistral
- Qwen (Alibaba)

Free to download; run on your own hardware; full control.

For research: choice depends on **budget**, **data sensitivity**, and **reproducibility** needs.

LLMs as Text Analysis Tools

(a) Zero-shot classification

Classify text **without any training data** — just describe the categories in a prompt.

Example: "Classify this tweet as positive, negative, or neutral."

(b) Summarisation

Condense long documents into concise summaries while preserving key information.

Example: "Summarise this parliamentary debate in three bullet points."

(c) Annotation / coding

Apply a structured **coding scheme** to text, replicating manual qualitative coding.

Example: "Code this paragraph for policy frames: economic, moral, or rights-based."

Prompt Engineering for Research

Key principles:

- ◎ **Be specific** — define the task, audience, and expected output
- 📖 **Provide context** — include background information and definitions
- 1 2 3 ☰ **Use examples** (few-shot) — show input–output pairs
- **</>** **Request structured output** — ask for JSON, CSV, or tables

Vague prompt

```
What is this text about?
```

Effective prompt

```
Classify this text as: economy,
health, or security. Return JSON:
{"label": "...", "conf": "..."}
```

Strengths and Limitations

Strengths

- **Flexible** — adapt via prompts
- **No training data** — zero/few-shot
- **Handles nuance** — context, irony
- **Multilingual** — many languages

Limitations

- **£ Cost** — API calls add up
- **Reproducibility** — stochastic
- **Hallucination** — confident but wrong
- **Black box** — hard to explain
- **Context limits** — finite window

Bottom line

LLMs are powerful but require **careful validation** against human judgements — especially for high-stakes research.

Comparing LLMs to Traditional Methods

	Dictionary	Topic models	Supervised	LLMs
Training data	None (lexicon)	None	Large labelled set	None (zero-shot)
Setup effort	Low	Medium	High	Low–medium
Nuance	Low	Medium	High	High
Reproducibility	Perfect	Variable	High	Low (stochastic)
Cost at scale	Negligible	Moderate	Low (post-train)	High (API)

Key message

LLMs **complement** traditional methods — they do not replace them. Use the right tool for the right task, and **validate** across approaches.

Next Up

Practical 4: LLM-Assisted Text Analysis

- Use an LLM API to classify text at scale
- Experiment with prompt design and few-shot examples
- Compare LLM annotations with human-coded labels
- Discuss reproducibility and validation strategies

Open the **Practical 4** notebook in Google Colab
(Python or R — your choice!)